Lab Sheet 3 — SQL Functions

```sql
USE coer_university;
DROP TABLE IF EXISTS Student;
CREATE TABLE Student (Student_ID INT PRIMARY KEY,Name VARCHAR(50),Gender VARCHAR(10),Age INT,Address
  VARCHAR(100),
Course VARCHAR(50),Percentage DECIMAL(5,2),PhoneNo VARCHAR(15),Dept_ID INT,Course_ID INT);
INSERT INTO Student VALUES
(101, 'Anshika', 'Female', 20, 'Jaipur', 'B.Tech CSE', 85.50, '9876543210', 1, 201),
(102, 'Radha', 'Female', 19, 'Kashipur', 'B.Tech CSE', 85.50, '9876543276', 1, 201),
(103, 'Jayshree', 'Female', 17, 'Jaipur', 'B.Tech AIML', 85.50, '9876543276', 1, 202),
(104, 'Ruhani', 'Female', 20, 'kashmir', 'B.Tech CSE', 85.50, '9876543276', NULL, 201),
(105, 'Shreya', 'Female', 20, 'Jaipur', 'B.Tech CSE', 85.50, '9876543276', 1, 201),
(106, 'Maya', 'Female', 20, 'rajasthan', 'B.Tech CSE', 85.50, '9876543276', NULL, 201),
(107, 'Rahul', 'Male', 21, 'Delhi', 'B.Tech CSE', 78.25, '9876543211', 1, 201),
(108, 'Priya', 'Female', 20, 'Mumbai', 'BCA', 91.00, '9876543212', 3, 204),
(109, 'Aman', 'Male', 22, 'Pune', 'B.Tech IT', 74.75, '9876543213', 2, 203),
(110, 'Neha', 'Female', 21, 'Kolkata', 'B.Tech CSE', 88.50, '9876543214', 1, 201),
(111, 'Rohit', 'Male', 20, 'Chandigarh', 'BCA', 69.25, '9876543215', 3, 204),
(112, 'Simran', 'Female', 22, 'Amritsar', 'B.Tech IT', 82.00, '9876543216', 2, 203);
SELECT * FROM Student;
```

Result Grid

Student_ID	Name	Gender	Age	Address	Course	Percentage	PhoneNo	Dept_ID	Course_ID
101	Anshika	Female	20	Jaipur	B.Tech CSE	85.50	9876543210	1	201
102	Radha	Female	19	Kashipur	B.Tech CSE	85.50	9876543276	1	201
103	Jayshree	Female	17	Jaipur	B.Tech AIML	85.50	9876543276	1	202
104	Ruhani	Female	20	kashmir	B.Tech CSE	85.50	9876543276	NULL	201
105	Shreya	Female	20	Jaipur	B.Tech CSE	85.50	9876543276	1	201
106	Maya	Female	20	rajasthan	B.Tech CSE	85.50	9876543276	NULL	201
107	Rahul	Male	21	Delhi	B.Tech CSE	78.25	9876543211	1	201
108	Priya	Female	20	Mumbai	BCA	91.00	9876543212	3	204
109	Aman	Male	22	Pune	B.Tech IT	74.75	9876543213	2	203
110	Neha	Female	21	Kolkata	B.Tech CSE	88.50	9876543214	1	201
111	Rohit	Male	20	Chandigarh	BCA	69.25	9876543215	3	204
112	Simran	Female	22	Amritsar	B.Tech IT	82.00	9876543216	2	203

```sql
DROP TABLE IF EXISTS Department;
CREATE TABLE Department (Dept_ID INT PRIMARY KEY, Dept_Name VARCHAR(50), Location VARCHAR(50));
INSERT INTO Department VALUES
(1, 'Computer Science', 'Jaipur'),
(2, 'Information Technology', 'Pune'),
(3, 'Computer Applications', 'Delhi'),
(4, 'Mechanical', 'Haridwar'),
(5, 'Civil', 'Meerut'),
(6, 'Human Resources', 'Roorkee');
SELECT * FROM Department;
```

Result Grid

Dept_ID	Dept_Name	Location
1	Computer Science	Jaipur
2	Information Technology	Pune
3	Computer Applications	Delhi
4	Mechanical	Haridwar
5	Civil	Meerut
6	Human Resources	Roorkee

```sql
DROP TABLE IF EXISTS Course;
CREATE TABLE Course (Course_ID INT PRIMARY KEY, Course_Name VARCHAR(50), Duration VARCHAR(20), Fees
  DECIMAL(10,2), Dept_ID INT);
INSERT INTO Course VALUES
(201, 'B.Tech CSE', '4 Years', 90000.00, 1),
(202, 'B.Tech AIML', '4 Years', 95000.00, 1),
(203, 'B.Tech IT', '4 Years', 88000.00, 2),
(204, 'BCA', '3 Years', 60000.00, 3),
(205, 'B.Tech Civil', '4 Years', 80000.00, 5);
SELECT * FROM Course;
```

Result Grid

Course_ID	Course_Name	Duration	Fees	Dept_ID
201	B.Tech CSE	4 Years	90000.00	1
202	B.Tech AIML	4 Years	95000.00	1
203	B.Tech IT	4 Years	88000.00	2
204	BCA	3 Years	60000.00	3
205	B.Tech Civil	4 Years	80000.00	5

```sql
DROP TABLE IF EXISTS Employee;
CREATE TABLE Employee (Emp_ID INT PRIMARY KEY, Emp_Name VARCHAR(50), Department VARCHAR(30), Salary
  DECIMAL(10,2),
Joining_Date DATE, Manager_ID INT, Dept_ID INT);
INSERT INTO Employee VALUES
(1, 'Aarav', 'CSE', 65000.00, '2022-06-01', NULL, 1),
(2, 'Diya', 'HR', 72000.50, '2023-01-15', 1, 6),
(3, 'Kabir', 'IT', 54000.00, '2021-09-10', 1, 2),
(4, 'Meera', 'Mechanical', 61000.00, '2020-03-20', 1, 4),
(5, 'Rohan', 'CSE', 48500.75, '2024-02-05', 1, 1),
(6, 'Sneha', 'IT', 58250.25, '2023-07-18', 3, 2),
(7, 'Vikram', 'CSE', 82000.00, '2019-11-25', NULL, 1),
(8, 'Pooja', NULL, 35000.00, '2024-08-12', 4, NULL);
SELECT * FROM Employee;
```

Result Grid

Emp_ID	Emp_Name	Department	Salary	Joining_Date	Manager_ID	Dept_ID
1	Aarav	CSE	65000.00	2022-06-01	NULL	1
2	Diya	HR	72000.50	2023-01-15	1	6
3	Kabir	IT	54000.00	2021-09-10	1	2
4	Meera	Mechanical	61000.00	2020-03-20	1	4
5	Rohan	CSE	48500.75	2024-02-05	1	1
6	Sneha	IT	58250.25	2023-07-18	3	2
7	Vikram	CSE	82000.00	2019-11-25	NULL	1
8	Pooja	NULL	35000.00	2024-08-12	4	NULL

```sql
DROP TABLE IF EXISTS Product;
CREATE TABLE Product (Product_ID INT PRIMARY KEY, Product_Name VARCHAR(50), Price DECIMAL(10,2), Quantity INT);
INSERT INTO Product VALUES
(1, 'Wooden Toy', 450.00, 10),
(2, 'Clay Diya', 80.00, 25),
(3, 'Handmade Vase', 350.00, 5),
(4, 'Jute Bag', 250.00, 12);
SELECT * FROM Product;
```

Result Grid

Product_ID	Product_Name	Price	Quantity
1	Wooden Toy	450.00	10
2	Clay Diya	80.00	25
3	Handmade Vase	350.00	5
4	Jute Bag	250.00	12

1. Demonstrate ABS() with suitable examples.

```sql
SELECT ABS(-25) AS Abs_Negative, ABS(25) AS Abs_Positive, ABS(-7.5) AS Abs_Decimal;
```

Result Grid

Abs_Negative	Abs_Positive	Abs_Decimal
25	25	7.5

2. Demonstrate ROUND() with suitable examples.

```sql
SELECT ROUND(123.4567, 2) AS Round_2, ROUND(123.4567, 0) AS Round_0, ROUND(123.4567, -1) AS Round_Tens;
```

Result Grid

Round_2	Round_0	Round_Tens
123.46	123	123

3. Demonstrate CEIL() and FLOOR().

```sql
SELECT CEIL(12.3) AS Ceiling_Value, FLOOR(12.8) AS Floor_Value, CEIL(-12.3) AS Ceil_Negative, FLOOR(-12.8) AS
  Floor_Negative;
```

Result Grid

Ceiling_Value	Floor_Value	Ceil_Negative	Floor_Negative
13	12	-12	-13

4. Demonstrate MOD() with suitable examples.

```
66  SELECT MOD(17, 5) AS Remainder, MOD(20, 4) AS Zero_Remainder, 17 % 5 AS Using_Percent;
```

Result Grid

Remainder	Zero_Remainder	Using_Percent
2	0	2

5. Calculate the rounded salary of employees.

```
67  SELECT Emp_Name, Salary, ROUND(Salary, 0) AS Rounded_Salary FROM Employee;
```

Result Grid

Emp_Name	Salary	Rounded_Salary
Aarav	65000.00	65000
Diya	72000.50	72001
Kabir	54000.00	54000
Meera	61000.00	61000
Rohan	48500.75	48501
Sneha	58250.25	58250
Vikram	82000.00	82000
Pooja	35000.00	35000

6. Find the total number of students using COUNT().

```
68  SELECT COUNT(*) AS Total_Students FROM Student;
```

Result Grid

Total_Students
12

7. Find the total salary of all employees using SUM().

```
69  SELECT SUM(Salary) AS Total_Salary FROM Employee;
```

Result Grid

Total_Salary
475751.50

8. Find the average salary using AVG().

```
70  SELECT AVG(Salary) AS Average_Salary FROM Employee;
```

Result Grid

Average_Salary
59468.937500

9. Find the highest salary using MAX().

```
71  SELECT MAX(Salary) AS Highest_Salary FROM Employee;
```

Result Grid

Highest_Salary
82000.00

10. Find the lowest salary using MIN().

```
72  SELECT MIN(Salary) AS Lowest_Salary FROM Employee;
```

Result Grid

Lowest_Salary
35000.00

11. Demonstrate UPPER() on student names.

```
73 • SELECT Name, UPPER(Name) AS Upper_Name FROM Student;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Name	Upper_Name
Anshika	ANSHIKA
Radha	RADHA
Jayshree	JAYSHREE
Ruhani	RUHANI
Shreya	SHREYA
Maya	MAYA
Rahul	RAHUL
Priya	PRIYA
Aman	AMAN
Neha	NEHA
Rohit	ROHIT
Simran	SIMRAN

12. Demonstrate LOWER() on student names.

```
74 • SELECT Name, LOWER(Name) AS Lower_Name FROM Student;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Name	Lower_Name
Anshika	anshika
Radha	radha
Jayshree	jayshree
Ruhani	ruhani
Shreya	shreya
Maya	maya
Rahul	rahul
Priya	priya
Aman	aman
Neha	neha
Rohit	rohit
Simran	simran

13. Demonstrate LENGTH() on student names.

```
75 • SELECT Name, LENGTH(Name) AS Name_Length FROM Student;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Name	Name_Length
Anshika	7
Radha	5
Jayshree	8
Ruhani	6
Shreya	6
Maya	4
Rahul	5
Priya	5
Aman	4
Neha	4
Rohit	5
Simran	6

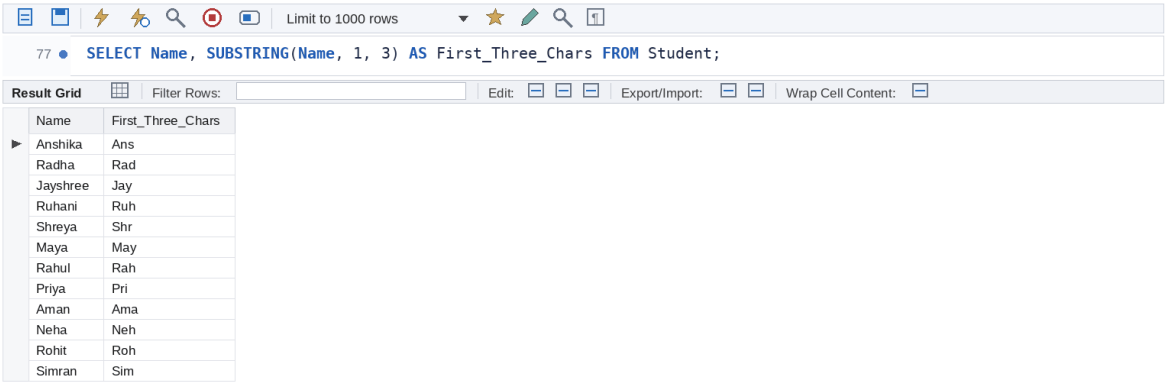
14. Demonstrate CONCAT() using first and last names.

```
76 • SELECT CONCAT('Anshika', ' ', 'Sharma') AS Full_Name;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Full_Name
Anshika Sharma

15. Demonstrate SUBSTRING()/SUBSTR() with suitable examples.

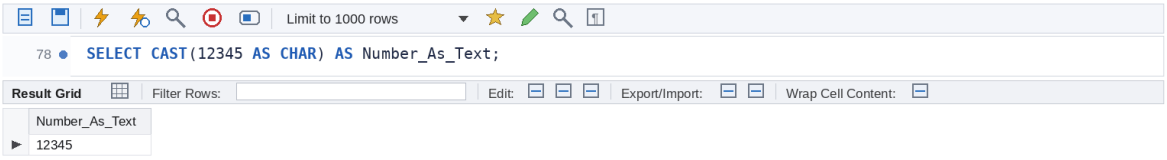


```
77  SELECT Name, SUBSTRING(Name, 1, 3) AS First_Three_Chars FROM Student;
```

Result Grid

Name	First_Three_Chars
Anshika	Ans
Radha	Rad
Jayshree	Jay
Ruhani	Ruh
Shreya	Shr
Maya	May
Rahul	Rah
Priya	Pri
Aman	Ama
Neha	Neh
Rohit	Roh
Simran	Sim

16. Convert a number into a character/string value.

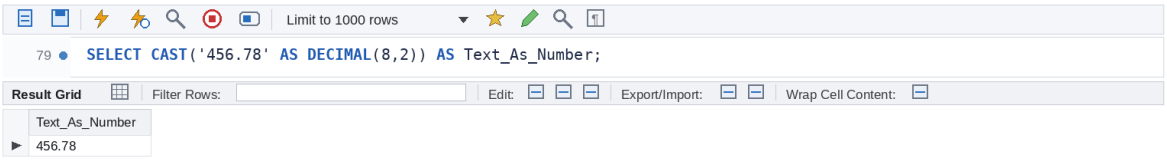


```
78  SELECT CAST(12345 AS CHAR) AS Number_As_Text;
```

Result Grid

Number_As_Text
12345

17. Convert a string into a numeric value.

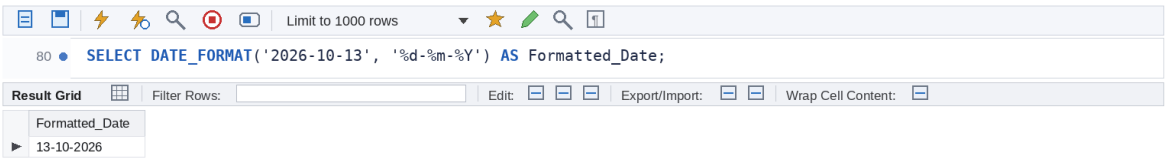


```
79  SELECT CAST('456.78' AS DECIMAL(8,2)) AS Text_As_Number;
```

Result Grid

Text_As_Number
456.78

18. Convert a date value into a different format.

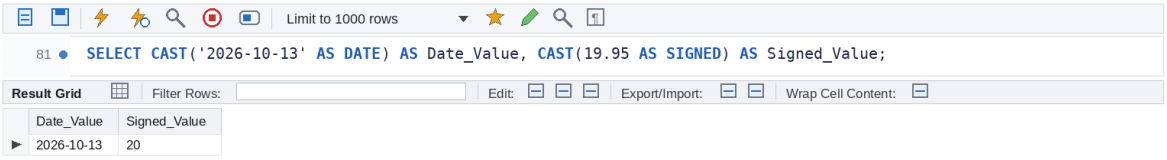


```
80  SELECT DATE_FORMAT('2026-10-13', '%d-%m-%Y') AS Formatted_Date;
```

Result Grid

Formatted_Date
13-10-2026

19. Demonstrate CAST() with suitable examples.

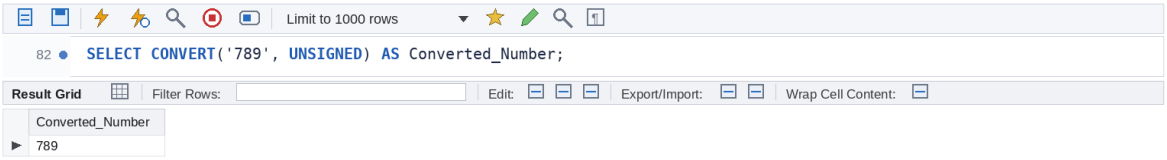


```
81  SELECT CAST('2026-10-13' AS DATE) AS Date_Value, CAST(19.95 AS SIGNED) AS Signed_Value;
```

Result Grid

Date_Value	Signed_Value
2026-10-13	20

20. Demonstrate CONVERT() where supported by the DBMS.

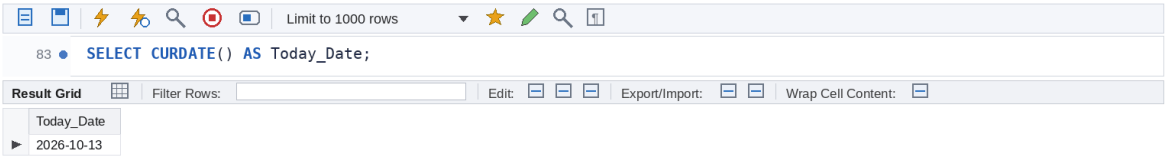


```
82  SELECT CONVERT('789', UNSIGNED) AS Converted_Number;
```

Result Grid

Converted_Number
789

21. Display the current date using a suitable date function.



```
83  SELECT CURDATE() AS Today_Date;
```

Result Grid

Today_Date
2026-10-13

22. Extract the year, month and day from a date.

```
84  SELECT YEAR('2024-08-17') AS Year_Part, MONTH('2024-08-17') AS Month_Part, DAY('2024-08-17') AS Day_Part;
```

Result Grid

Year_Part	Month_Part	Day_Part
2024	8	17

23. Find the difference between two dates.

```
85  SELECT DATEDIFF('2026-10-13', '2026-10-01') AS Days_Difference;
```

Result Grid

Days_Difference
12

24. Add a specified number of days to a date.

```
86  SELECT CURDATE() AS Today_Date, DATE_ADD(CURDATE(), INTERVAL 10 DAY) AS After_10_Days;
```

Result Grid

Today_Date	After_10_Days
2026-10-13	2026-10-23

25. Display employees' joining dates in a required date format.

```
87  SELECT Emp_Name, Joining_Date, DATE_FORMAT(Joining_Date, '%d %b %Y') AS Formatted_Joining_Date FROM Employee;
```

Result Grid

Emp_Name	Joining_Date	Formatted_Joining_Date
Aarav	2022-06-01	01 Jun 2022
Diya	2023-01-15	15 Jan 2023
Kabir	2021-09-10	10 Sep 2021
Meera	2020-03-20	20 Mar 2020
Rohan	2024-02-05	05 Feb 2024
Sneha	2023-07-18	18 Jul 2023
Vikram	2019-11-25	25 Nov 2019
Pooja	2024-08-12	12 Aug 2024